

**Human-Computer Interaction**

# **Accessibility, Aging & AI**

**Professor Bilge Mutlu**

# Today's Agenda

- Topic overview: *Accessibility, Aging & AI*
- The frame: *ability-based design* → *adaptive AI mediation*
- Provocation: *AI-as-liberation or AI-as-erasure?*
- Discussion

# This term's frame

*Enduring fundamental → How AI is changing it → Depth (a claim to argue with)*

- **Fundamental:** design for the full range of human ability; access is *relational*
- **How AI changes it:** personalized **mediation** becomes possible, but the same systems can **encode ableism**
- **Depth: Deep:** accommodation shifts from fixed fixes to *adaptive mediation*

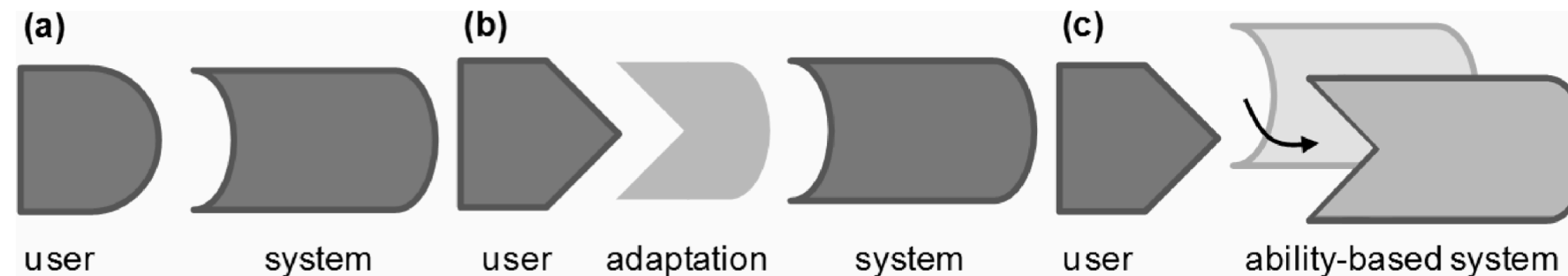
# This week's readings

- **Required**
  - Wobbrock et al., *Ability-Based Design* (TACCESS 2011)
  - Glazko et al., *An Autoethnographic Case Study of GenAI's Utility for Accessibility* (ASSETS 2023)
- **Optional**: Morris, *AI and Accessibility* (CACM 2020) · Whittaker, Morris et al., *Disability, Bias, and AI* (AI Now, 2019)

# **Topic overview:** ***Accessibility, Aging &*** ***AI***

# Ability-based design<sup>1</sup>

- **Ability:** "focus on ability not dis-ability, striving to leverage all that users can do"
- **Accountability:** "respond to poor performance by changing systems, not users"



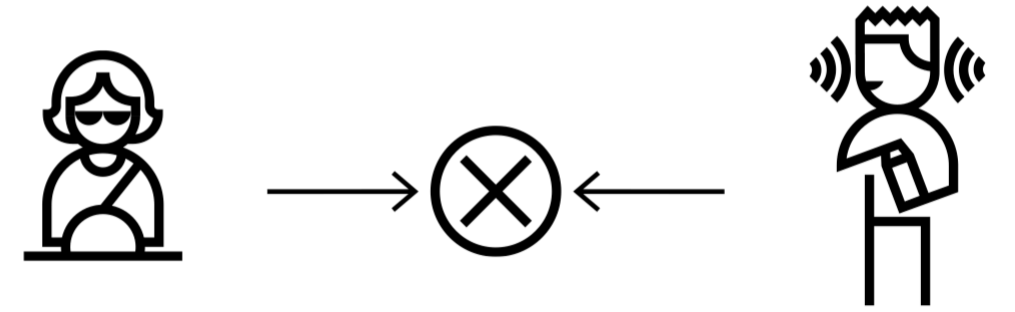
<sup>1</sup>Wobbrock, Kane, Gajos, Harada & Froehlich (2011). Ability-Based Design: Concept, Principles and Examples. *ACM TACCESS*, 3(3), Fig. 2 and Table 1.

# Access is relational<sup>2</sup>

Disability "results from the interaction between health conditions... and a range of environmental and personal factors."

Change the context and the barrier can disappear: the design carries the accommodation.

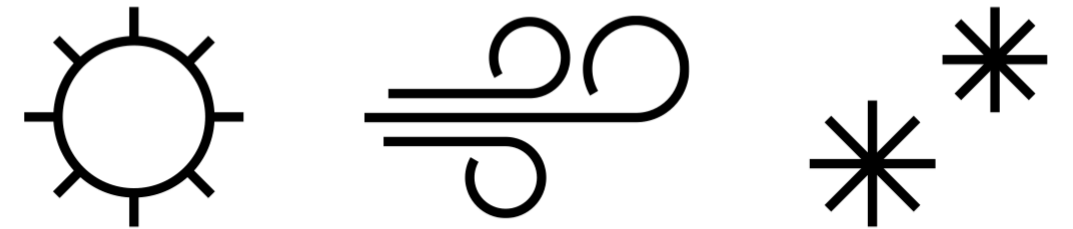
Between  
humans



Can't type

Can't hear

Human+  
enviroment

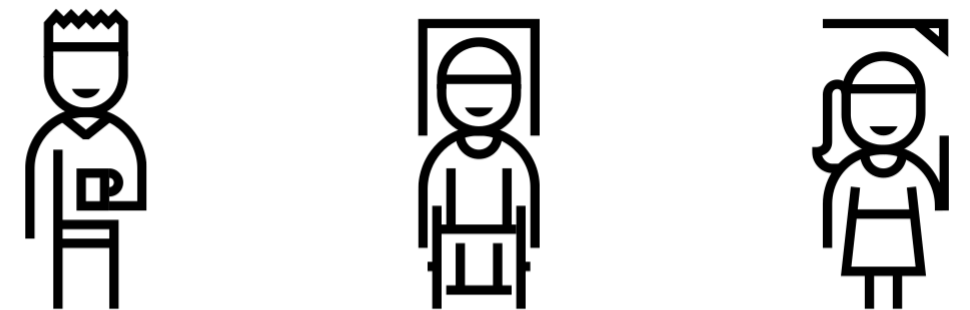


Glare from sun

Windy

Cold

Human+  
object



Left-handed user

Narrow door

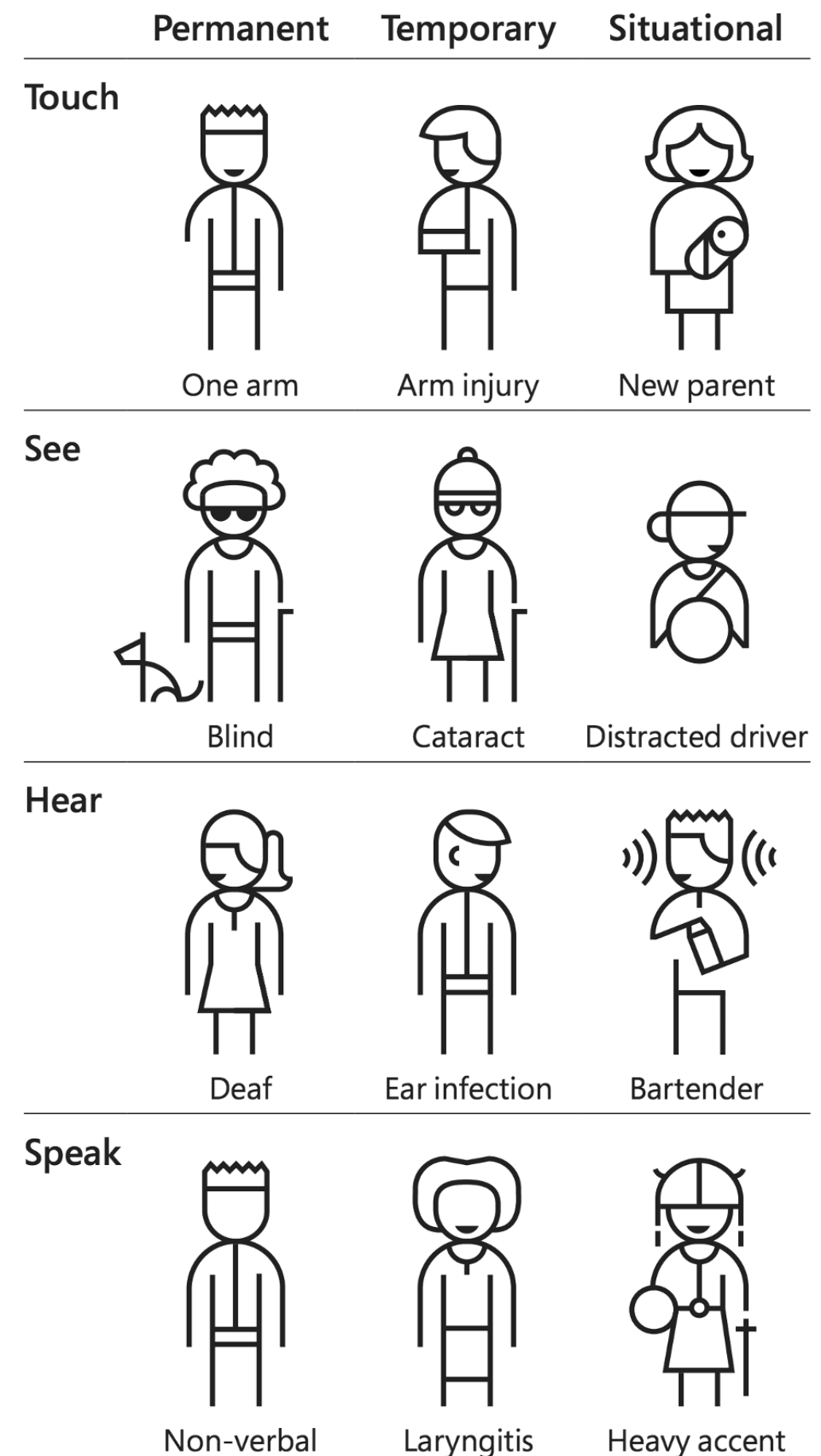
Tall shelf

<sup>2</sup>WHO (2023). *Disability* fact sheet. Figure: Microsoft Inclusive Design Toolkit.

# Ability changes over time<sup>3</sup>

- **1.3 billion** people, "1 in 6 of us," experience significant disability
- The number is growing, in part because of **people living longer**
- **Temporary** and **situational** impairments touch everyone

<sup>3</sup>WHO (2023). *Disability* fact sheet. Figure: Microsoft Inclusive Design Toolkit (persona spectrum).



# What AI changes: personalized mediation

On-demand, bespoke adaptation is finally possible.

- Multimodal **description** and scene understanding
- Real-time **captioning** and translation
- **Simplification** and reading support
- **Agentic** mediation that acts on the user's behalf

👉 Accommodation moves from *fixed add-ons* to *adaptive mediation*.

# The duality, first-person<sup>4</sup>

A three-month autoethnography by researchers **with and without** disabilities.

- **Useful:** references, social messages, drafting, code for a blind developer
- **But** it "almost always required human verification"
- It would "repeatedly provide a false solution" while "stating confidently that it understood accessibility guidelines"

---

<sup>4</sup>Glazko, Yamagami, Desai, Mack, Potluri, Xu & Mankoff (2023). An Autoethnographic Case Study of Generative AI's Utility for Accessibility. *ASSETS '23*.

# AI can encode ableism<sup>5</sup> <sup>6</sup>

The same models that mediate also **allocate** and **represent**.

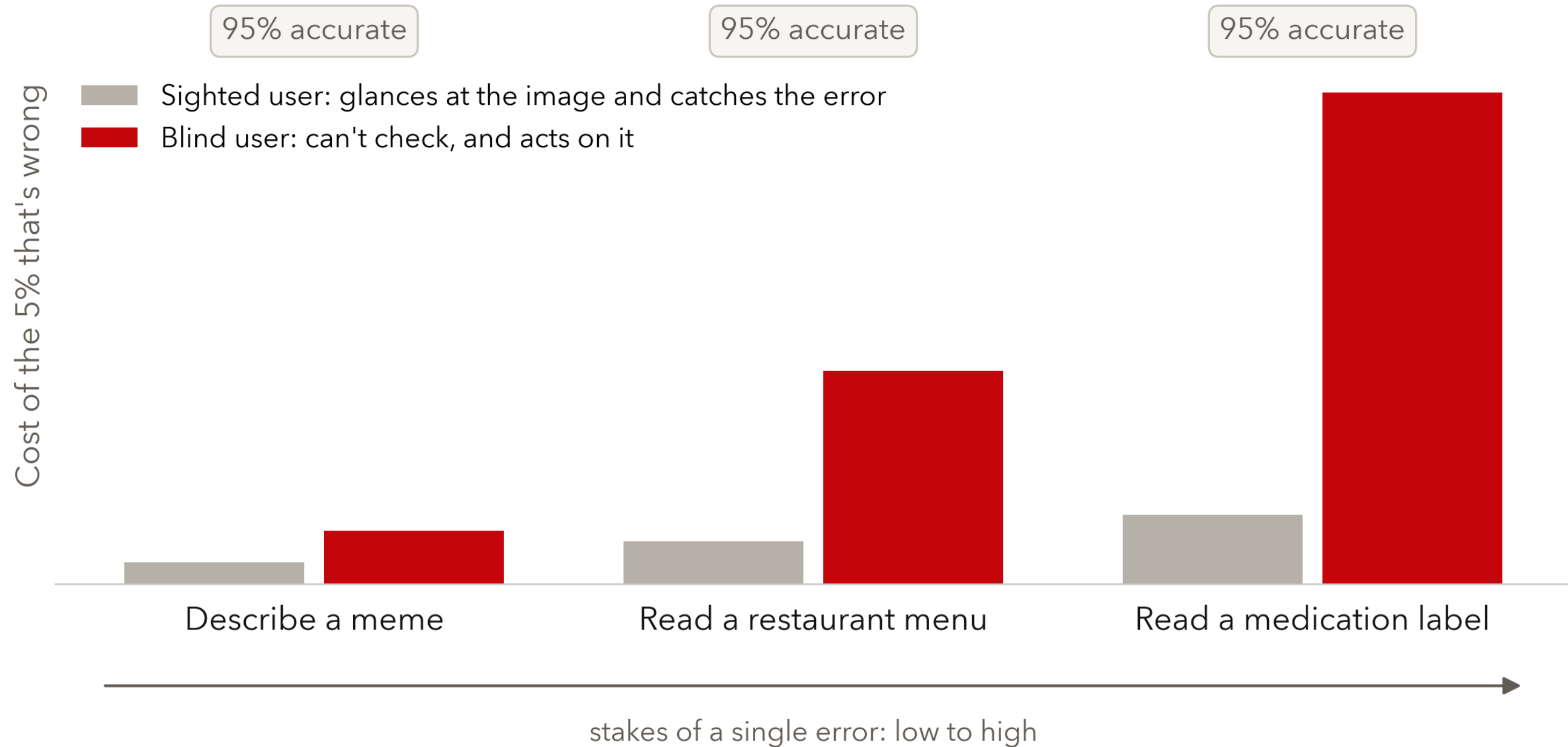
- GPT-4 ranked résumés **lower** when they listed disability-related honors<sup>5</sup>
- Systems trained on ableist data reproduce it at scale<sup>6</sup>
- "Better tools" and "new harm" are **not** separate stories

---

<sup>5</sup> Glazko, Mohammed, Kosa, Potluri & Mankoff (2024). Identifying and Improving Disability Bias in GPT-Based Resume Screening. *FAccT '24*.

<sup>6</sup> Whittaker, Morris et al. (2019). *Disability, Bias, and AI*. AI Now Institute. (optional)

## The failure-cost asymmetry: same accuracy, different cost of being wrong



Conceptual figure: illustrative values.

**Ask who bears the cost of being wrong.**

# Fold-in: participatory & justice design<sup>7</sup>

Not just an attribute of the studies we cite, but a **method**.

- Scandinavian **participatory design**: those affected are co-designers
- **Co-design** and **design justice**: shift who holds design power
- The operational form of "*nothing about us without us*"

---

<sup>7</sup> Costanza-Chock, S. (2020). *Design Justice: Community-Led Practices to Build the Worlds We Need*. MIT Press. Open access: [designjustice.mitpress.mit.edu](https://designjustice.mitpress.mit.edu)

# Guardrails coda: whose descriptions count?

- **Data:** models learn from data that may not include you: "without better training data, these tasks are likely to remain inaccessible even as GAI quality advances"
  - **Output:** GPT-4 marked down résumés with disability-related honors, and a draft "replaced nuanced arguments with simplified, ableist tropes"
- 👉 Was it built *with* disabled people, or only *for* them?

# Can the user check it? Can they undo it?

- **Verify:** tools should "facilitate verification rather than simply proclaiming success"
  - **The risk:** a less experienced user "could incorporate recommendations under the false belief that they solved an accessibility issue"
  - **Control:** users need the means to "inspect, override, discard, revert" what the system adapts
- 👉 Where errors are costly, the uncertainty has to be visible.

# A design test for AI access tools

1. **Built with** disabled people, not only for them
  2. **Says what it doesn't know**, especially where errors are costly
  3. Lets the user **inspect, override, and undo**
  4. **Changes the system, not the user**
- 👉 Does it extend the user's agency, or replace it?

# **Liberation or *erasure*?**

# The tension, today

- **Liberation**: bespoke access at scale
  - **Erasure**: ableism at scale, and user agency displaced
  - **Wobbrock**: change the system, not the user
  - **Glazko**: useful, but only if you can verify it
- 👉 The test: does the system *center* disabled users, or *speak over* them?

# Your provocation (*post one slide*)

Pick one:

1. **Provocation**: when does personalized AI mediation *empower* disabled users, and when does it *displace* their agency?
2. **Critical artifact**: an AI accessibility tool or output that helps *and* misleads (an auto-caption, a scene description)
3. **Design / policy**: one principle for "*nothing about us without us*" in AI-assistive design

# Discussion

# Today's roles

**Reporter**

**Skeptic**

**Connector**

---

**Seat 5**

**Seat 6**

**Seat 1**

Same in every pod. **Seats 2, 3 and 4** have no role today; you are still in the argument.

# Discussion Format

**In your pod, on your pod monitor:** ~30 minutes.

- **Everyone shows their slide:** about 2 minutes each, cast from your own laptop
- **Skeptic** puts a counterargument on the table · **Connector** ties it to an earlier week
- **Reporter** writes the pod's three lines (agreed, split, unresolved) and delivers them to the room

# What's Next?

- **Friday: Method-Design Crit (M6)**
  - **Due 9:30 am:** Step-by-Step Experimental Design (*individual*); bring it, it's what you crit
  - **If you can:** LFH **Ch. 13** (Measuring the human) · optional Ch. 5 (Surveys)